LYAPUNOV BRANCH EXCLUSIONS FOR ANCIENT NAVIER–STOKES FLOWS

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ABSTRACT. We isolate the parts of a Lyapunov-based ancient-solution analysis for the three-dimensional Navier–Stokes equations which presently give rigorous branch exclusions. For the unforced equations, bounded ancient axisymmetric subclasses are eliminated by known Liouville theorems: no swirl, pointwise scale-invariant control, finite L^p control of the swirl variable $\Gamma = ru_\theta$, and axial periodicity with bounded Γ. A separate perturbative weighted-vorticity branch is excluded by a semiflow Lyapunov functional built from Gallay–Wayne's small-data decay theorem in weighted vorticity spaces. For the strained Burgers-vortex model, the Gallay–Maekawa stability theorem yields a profile-relative Lyapunov functional and an ancient rigidity theorem near the Burgers-vortex family. Finally, in the open bounded-swirl axisymmetric case, we prove the exact scalar Γ-defect identity and the anisotropic shell-error decomposition. In the whole-space smooth bounded setting, the clean global identity is centered at the only constant compatible with regularity on the axis, namely $\Gamma = 0$. This identity does not close the open branch by itself; it gives a precise coercive shell-control criterion under which the branch would reduce to vanishing Γ.

1. INTRODUCTION

We consider ancient solutions of the incompressible Navier–Stokes equations on $\mathbb{R}^3$,

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \Delta u, \qquad \nabla \cdot u = 0, \qquad t < 0. \tag{1.1}$$

The purpose of this paper is not to claim a new Liouville theorem for all bounded ancient solutions. Rather, it records exactly which branches are closed by existing Liouville theorems or by a genuine Lyapunov construction, and it writes the remaining bounded-swirl obstruction as an explicit shell-transport problem.

There are three distinct mechanisms below.

First, several axisymmetric branches of the unforced equations are already closed by known Liouville theorems. If a bounded ancient mild solution is axisymmetric without swirl, Koch–Nadirashvili–Seregin–Sverak imply that it is Galilean-trivial. The same conclusion holds under the pointwise scale-invariant bound $|u| \le C/r$. Two swirl-bearing subclasses are also closed: Lei–Zhang–Zhao prove triviality when $\Gamma = ru_\theta \in L^\infty_t L^p_x$, $1 \le p < \infty$, and Lei–Ren–Zhang prove triviality for z-periodic bounded ancient axisymmetric solutions with bounded Γ.

Second, a perturbative weighted-vorticity branch is closed by an actual Lyapunov functional. Gallay–Wayne's small-data theory for the rescaled three-dimensional vorticity equation gives exponential decay in weighted $L^2(m)$ spaces. From this decay one defines

$$\mathcal{L}(w_0) = \sup_{\tau \ge 0} e^{2\tau} \|\Phi_\tau w_0\|^2_{L^2(m)}.$$

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This functional is coercive on the small ball and contracts along the forward semiflow. A bounded ancient trajectory contained in that ball is therefore forced to vanish by letting the initial time tend to $-\infty$.

Third, the same Lyapunov construction closes a model structured-vortex branch for the strained Burgers-vortex equation. The neutral circulation parameter is removed by horizontal integration of the vertical vorticity; Gallay–Maekawa's stability theorem then provides exponential decay on the fixed-circulation subspace. The associated profile-relative functional excludes all ancient solutions which remain in the small weighted tube around the Burgers-vortex family. This is a theorem for the strained model. Importing it into an unforced Navier–Stokes blow-up analysis requires an additional analytic reduction to that model.

The last part treats the open bounded-swirl axisymmetric branch. The correct first Lyapunov variable is not the full velocity defect, but the scalar swirl quantity

$$\Gamma = ru_\theta.$$

For $W = \Gamma$ and for any nonnegative axisymmetric test weight ϕ, we prove

$$\frac{1}{2}\frac{d}{dt}\int_{\mathbb{R}^3} W^2\phi\,dx + \int_{\mathbb{R}^3} |\nabla W|^2\phi\,dx = \frac{1}{2}\int_{\mathbb{R}^3} W^2\left(\partial_t\phi + \Delta\phi + b\cdot\nabla\phi + \frac{2}{r}\partial_r\phi\right)dx, \tag{1.2}$$

where $b = u_r e_r + u_z e_z$. For $W = \Gamma - c$, the same formula has an additional axis boundary term unless $c = 0$ or the test weight avoids the axis. With $\phi(r, z, t) = \lambda_R(r)\eta_Z(z - \zeta(t))$, this becomes an exact anisotropic shell formula whose only error channels are radial shell transport, axial drift mismatch, and cutoff curvature. The radial diffusion operator is $\partial_r^2 + 3r^{-1}\partial_r$, so the adapted radial barrier is harmonic for the four-dimensional radial Laplacian, not an isotropic Gaussian.

The identity (1.2) is rigorous and useful, but it does not prove the full bounded-Γ Liouville conjecture. We therefore state the precise coercive shell-control criterion that would force Γ to vanish in expanding windows. The open analytic point is to verify that criterion for every bounded ancient axisymmetric solution with bounded Γ and no decay or periodicity.

Relation with the literature. Bounded ancient solutions enter the Navier–Stokes regularity theory through blow-up and compactness arguments. The backward-uniqueness method of Escauriaza–Seregin–Sverak [3] proves regularity for $L^\infty_t L^3_x$ solutions and is one of the standard mechanisms by which critical control rules out singularity formation. In the axisymmetric setting, Koch–Nadirashvili–Seregin–Sverak [7] developed Liouville theorems for bounded ancient solutions and connected them to lower bounds for axisymmetric singularity formation. The related quantitative lower-bound program for axisymmetric solutions with swirl was advanced by Chen–Strain–Tsai–Yau [1, 2], while Seregin–Sverak [12] treated Type I local axisymmetric singularities.

The finite-L^p swirl branch used here is the bounded ancient mild theorem of Lei–Zhang–Zhao [10]. It should be distinguished from the earlier Lei–Zhang theorem [8], which proves regularity and Liouville consequences under scale-invariant control of the meridional drift $b = u_r e_r + u_z e_z$, for instance $b \in L^\infty_t \mathrm{BMO}^{-1}_x$. The periodic bounded-swirl branch is the theorem of Lei–Ren–Zhang [9] on $\mathbb{R}^2 \times \mathbb{T}^1$. These results are used below only in the precise forms stated in Theorems 3.1 to 3.4; the paper does not interpolate between them or assert an unstated Liouville theorem for the whole nonperiodic bounded-Γ class.

The Lyapunov arguments in the perturbative model sections rely on stability theory rather than on axisymmetric Liouville theorems. Gallay–Wayne [4] established decay for the rescaled vorticity equation in weighted spaces under smallness and algebraic spatial decay. For Burgers vortices, linear and low-Reynolds-number stability results such as Schmid–Rossi [11] and Gallay–Wayne [5] precede the nonlinear three-dimensional stability theorem of Gallay–Maekawa [6], which is the stability input used in Theorem 5.4. The Burgers-vortex conclusions in this paper are therefore model conclusions unless an independent reduction from an unforced Navier–Stokes regime to the strained equation has been proved.

2. Preliminaries

2.1. Ancient solutions and Galilean equivalence.

Definition 2.1. A velocity-pressure pair (u, p) is a bounded ancient mild solution of (1.1) if it solves (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$ in the mild sense on every finite time interval, namely for every $-\infty < s < t < 0$,

$$u(t) = e^{(t-s)\Delta}u(s) - \int_s^t e^{(t-\sigma)\Delta}\mathbb{P}\nabla \cdot (u(\sigma) \otimes u(\sigma))\, d\sigma \tag{2.1}$$

in distributions, where $\mathbb{P}$ is the Leray projector, and

$$\|u\|_{L^\infty(\mathbb{R}^3 \times (-\infty,0))} < \infty.$$

The pressure is understood modulo functions of time.

Remark 2.2 (Regularity used below). The branch exclusions in Theorems 3.5 to 3.8 use the cited Liouville theorems in their stated bounded ancient mild or weak classes. The differential identities in Theorems 6.1, 6.2 and 6.4 are proved for smooth axisymmetric solutions. When they are invoked for bounded ancient mild axisymmetric solutions, the required additional input is that the solution is smooth on $\mathbb{R}^3 \times (-\infty, 0)$ and that Γ, $\nabla\Gamma$, and the displayed weighted integrals are finite for the chosen compactly supported weights. These are included explicitly in the shell-control assumption below, rather than inferred silently.

Lemma 2.3 (Galilean reduction)**.** *Let (u, p) solve (1.1). For every constant vector $c \in \mathbb{R}^3$,*

$$\widetilde{u}(x, t) = u(x + ct, t) - c, \qquad \widetilde{p}(x, t) = p(x + ct, t)$$

also solves (1.1). In particular, every constant velocity field is Galilean-equivalent to zero.

Proof. All derivatives below are evaluated at the point $(x + ct, t)$. By the chain rule,

$$\partial_t\widetilde{u} = (\partial_t u + c \cdot \nabla u)(x + ct, t), \qquad (\widetilde{u} \cdot \nabla)\widetilde{u} = ((u - c) \cdot \nabla u)(x + ct, t).$$

Adding these two identities gives

$$\partial_t\widetilde{u} + (\widetilde{u} \cdot \nabla)\widetilde{u} = (\partial_t u + (u \cdot \nabla)u)(x + ct, t).$$

Also

$$\Delta\widetilde{u}(x, t) = (\Delta u)(x + ct, t), \qquad \nabla\widetilde{p}(x, t) = (\nabla p)(x + ct, t),$$

and

$$\nabla \cdot \widetilde{u}(x, t) = (\nabla \cdot u)(x + ct, t).$$

Substituting the equation for (u, p) at $(x + ct, t)$ therefore gives

$$\partial_t\widetilde{u} + (\widetilde{u} \cdot \nabla)\widetilde{u} + \nabla\widetilde{p} = \Delta\widetilde{u}, \qquad \nabla \cdot \widetilde{u} = 0.$$

If $u \equiv c$, then $\widetilde{u}(x, t) = c - c = 0$ for every (x, t). $\square$

2.2. **Axisymmetric notation.** Write cylindrical coordinates as

$$x = (r\cos\theta, r\sin\theta, z), \qquad u = u_r e_r + u_\theta e_\theta + u_z e_z.$$

The solution is axisymmetric if u_r, u_θ, u_z are independent of θ. It is swirl-free if $u_\theta \equiv 0$. The scale-invariant swirl variable is

$$\Gamma = r u_\theta.$$

For axisymmetric solutions the meridional drift is

$$b = u_r e_r + u_z e_z.$$

3. Exact axisymmetric Liouville branch exclusions

The results in this section are Liouville imports for the actual unforced Navier–Stokes equations. They are stronger than model stability statements: no reduction to an auxiliary equation is involved.

3.1. **External Liouville inputs.** The following results are used as external theorems. They are stated here in the precise forms needed in the sequel.

Theorem 3.1 (Koch–Nadirashvili–Seregin–Sverak, no swirl)**.** *Let u be a bounded ancient weak axisymmetric solution of* (1.1) *on $\mathbb{R}^3 \times (-\infty, 0)$. If $u_\theta \equiv 0$, then*

$$u(x,t) = b(t)e_z$$

for some bounded scalar function $b(t)$.

Proof. This is the no-swirl Liouville theorem of Koch–Nadirashvili–Seregin–Sverak [7], stated only in the form used below. □

Theorem 3.2 (Koch–Nadirashvili–Seregin–Sverak, pointwise branch)**.** *Let u be a bounded ancient weak axisymmetric solution of* (1.1) *on $\mathbb{R}^3 \times (-\infty, 0)$. If*

$$|u(x,t)| \le \frac{C}{r} \qquad (r > 0,\ t < 0),$$

then $u \equiv 0$.

Proof. This is Theorem 5.3 of [7]. □

Theorem 3.3 (Lei–Zhang–Zhao)**.** *Let u be a bounded ancient mild axisymmetric solution of* (1.1) *on $\mathbb{R}^3 \times (-\infty, 0)$. If, for some $1 \le p < \infty$,*

$$\Gamma = r u_\theta \in L_t^\infty L_x^p\big(\mathbb{R}^3 \times (-\infty, 0)\big),$$

then u is a constant vector field.

Proof. This is the ancient axisymmetric L^p-swirl Liouville theorem of Lei–Zhang–Zhao [10], stated in the form used here. □

Theorem 3.4 (Lei–Ren–Zhang)**.** *Let u be a bounded ancient mild axisymmetric solution of* (1.1) *on $\mathbb{R}^2 \times \mathbb{T}^1$. If $\Gamma = r u_\theta$ is bounded, then*

$$u(x,t) = c e_z$$

for some constant $c \in \mathbb{R}$.

Proof. This is the periodic bounded-swirl Liouville theorem of Lei–Ren–Zhang [9]. □

3.2. Consequences for branch exclusions.

Theorem 3.5 (No-swirl ancient branch)**.** *Let u be a bounded ancient mild solution of* (1.1)*. If u is axisymmetric and $u_\theta \equiv 0$, then u is Galilean-trivial. More precisely, after rotation of the notation around the symmetry axis,*

$$u(x,t) = ce_z \qquad \textit{for some } c \in \mathbb{R}.$$

Proof. Theorem 3.1 gives $u(x,t) = b(t)e_z$. This means that u is independent of x and points along the symmetry axis. Hence $\Delta u = 0$, and $(u \cdot \nabla)u = 0$. Substitution into (1.1) gives

$$b'(t)e_z + \nabla p(x,t) = 0.$$

The preceding identity alone allows the pressure to absorb $b'(t)e_z$. The mild formulation fixes the velocity more strongly. For any $-\infty < s < t < 0$, (2.1) gives

$$u(t) = e^{(t-s)\Delta}u(s) - \int_s^t e^{(t-\sigma)\Delta}\mathbb{P}\nabla \cdot (u(\sigma) \otimes u(\sigma))\, d\sigma.$$

Here $u(s) = b(s)e_z$ is a constant vector field, so

$$e^{(t-s)\Delta}u(s) = u(s) = b(s)e_z.$$

Moreover $u(\sigma) \otimes u(\sigma)$ is a constant matrix in x, so $\nabla \cdot (u(\sigma) \otimes u(\sigma)) = 0$ distributionally. The integral term is therefore zero. Thus $b(t)e_z = b(s)e_z$ for every $s < t < 0$, and b is constant. The resulting constant velocity field is removed by Theorem 2.3. ☐

Theorem 3.6 (Pointwise scale-invariant axisymmetric branch)**.** *Let u be a bounded ancient mild axisymmetric solution of* (1.1)*. If*

$$|u(x,t)| \le \frac{C}{r} \qquad (r > 0,\ t < 0),$$

then $u \equiv 0$.

Proof. The hypothesis makes u a bounded ancient axisymmetric weak solution satisfying the pointwise scale-invariant condition in Theorem 3.2. A mild solution is distributional, and boundedness gives the local integrability required in the weak formulation. Applying Theorem 3.2 gives $u \equiv 0$. ☐

Theorem 3.7 (L^p-controlled swirl branch)**.** *Let u be a bounded ancient mild axisymmetric solution of* (1.1)*. Assume that for some $1 \le p < \infty$,*

$$\Gamma = ru_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Then u is Galilean-trivial.

Proof. If $u_\theta \equiv 0$, then u lies in the no-swirl class and Theorem 3.5 gives Galilean triviality. If $u_\theta \not\equiv 0$, then the displayed assumption is the finite-L^p swirl hypothesis in Theorem 3.3. Therefore u is a constant vector field. By Theorem 2.3, every constant vector field is equivalent to zero under a Galilean transform. ☐

Theorem 3.8 (Periodic bounded-swirl branch)**.** *Let u be a bounded ancient mild axisymmetric solution of* (1.1)*. Assume that u is periodic in z and*

$$\Gamma = ru_\theta \in L^\infty\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Then $u(x,t) = ce_z$ for some $c \in \mathbb{R}$. Hence u is Galilean-trivial.

Proof. Let $Z_0 > 0$ be an axial period. Since $u(r, \theta, z + Z_0, t) = u(r, \theta, z, t)$, the solution descends to the quotient $\mathbb{R}^2 \times (\mathbb{R}/Z_0\mathbb{Z})$. The theorem of Lei–Ren–Zhang is invariant under the Navier–Stokes scaling

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \qquad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t).$$

Choosing $\lambda = Z_0/(2\pi)$ normalizes the axial period to 2π, so the quotient is $\mathbb{R}^2 \times \mathbb{T}^1$. Boundedness of u is preserved up to the constant factor λ, and

$$r(u_\lambda)_\theta(r, z, t) = r\lambda u_\theta(\lambda r, \lambda z, \lambda^2 t) = (\lambda r) u_\theta(\lambda r, \lambda z, \lambda^2 t),$$

so boundedness of $\Gamma = r u_\theta$ is preserved with the same L^∞ bound under this scaling. Theorem 3.4 then gives $u(x, t) = c e_z$ for some constant c. Theorem 2.3 removes this constant axial drift. □

Corollary 3.9 (Closed unforced axisymmetric branches)**.** *Let $\mathcal{A}$ be any family of bounded ancient mild axisymmetric solutions of* (1.1)*. Suppose every $u \in \mathcal{A}$ satisfies at least one of the following conditions:*

(i) $u_\theta \equiv 0$*;*
(ii) $|u(x, t)| \le C/r$*;*
(iii) $r u_\theta \in L^\infty_t L^p_x$ *for some* $1 \le p < \infty$*;*
(iv) *u is periodic in z and* $r u_\theta \in L^\infty$*.*

Then every element of $\mathcal{A}$ is Galilean-trivial. If the class is studied in a gauge where constant drifts are fixed to zero, then $\mathcal{A} = \{0\}$.

Proof. Fix $u \in \mathcal{A}$. By hypothesis, u satisfies at least one of the four listed alternatives. In case (i), Theorem 3.5 applies. In case (ii), Theorem 3.6 applies. In case (iii), Theorem 3.7 applies. In case (iv), Theorem 3.8 applies. Each conclusion is Galilean triviality, and the extra gauge assumption converts Galilean triviality into the equality $u = 0$. □

4. A perturbative weighted-vorticity Lyapunov exclusion

We next record a genuine Lyapunov exclusion for a perturbative weighted branch. This section is independent of axisymmetry.

4.1. **Rescaled vorticity equation.** Let $w = \nabla \times v$ be the vorticity in forward self-similar variables. The Gallay–Wayne rescaled vorticity equation is

$$\partial_\tau w = \Lambda w - (v \cdot \nabla) w + (w \cdot \nabla) v, \qquad \nabla \cdot w = 0, \tag{4.1}$$

where

$$\Lambda w = \Delta w + \frac{1}{2} \xi \cdot \nabla w + w, \tag{4.2}$$

and v is recovered from w by the three-dimensional Biot–Savart law. For $m \ge 0$, set

$$L^2_\sigma(m) = \left\{ f \in L^2(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot f = 0, \quad \|f\|_m < \infty \right\},$$

where

$$\|f\|_m^2 = \int_{\mathbb{R}^3} (1 + |\xi|)^{2m} |f(\xi)|^2 \, d\xi.$$

Theorem 4.1 (Gallay–Wayne small-data decay)**.** *Let $0 < \mu \le 1$ and $m > 2\mu + \frac{1}{2}$. There exist constants $\rho_m > 0$ and $C_m \ge 1$ such that if $\|w_0\|_m \le \rho_m$, then* (4.1) *has a unique global mild solution $w(\tau) = \Phi_\tau w_0$ in*

$$C([0, \infty); L^2_\sigma(m))$$

and

$$\|\Phi_\tau w_0\|_m \le C_m e^{-\mu \tau} \|w_0\|_m, \qquad \tau \ge 0.$$

In particular, if $m > \frac{7}{2}$, one may use $\mu = 1$.

Proof. This is the Gallay–Wayne weighted small-data theorem for the rescaled three-dimensional vorticity equation [4]. It is used as an external input. The only conclusions used below are existence, uniqueness, the semiflow property induced by uniqueness, and the displayed exponential decay estimate. □

Definition 4.2 (Perturbative weighted ancient branch)**.** Fix $m > \frac{7}{2}$, and let ρ_m be the small-data radius in Theorem 4.1 with $\mu = 1$. The perturbative weighted ancient branch is the set of all

$$w \in C(\mathbb{R}; L^2_\sigma(m))$$

which solve (4.1) for all $\tau \in \mathbb{R}$ and satisfy

$$\sup_{\tau \in \mathbb{R}} \|w(\tau)\|_m \leq \rho_m.$$

Solving for all $\tau \in \mathbb{R}$ means that for every $s \in \mathbb{R}$, the shifted function $w_s(\theta) = w(s + \theta)$, $\theta \geq 0$, is the Gallay–Wayne mild solution with initial datum $w(s)$.

Proposition 4.3 (Semiflow Lyapunov functional)**.** *Fix $m > \frac{7}{2}$. For $\|w_0\|_m \leq \rho_m$, define*

$$\mathcal{L}_m(w_0) = \sup_{\tau \geq 0} e^{2\tau} \|\Phi_\tau w_0\|_m^2. \tag{4.3}$$

Then $\mathcal{L}_m$ is well-defined and satisfies

$$\|w_0\|_m^2 \leq \mathcal{L}_m(w_0) \leq C_m^2 \|w_0\|_m^2. \tag{4.4}$$

Moreover, for every $s \geq 0$,

$$\mathcal{L}_m(\Phi_s w_0) \leq e^{-2s} \mathcal{L}_m(w_0). \tag{4.5}$$

Proof. By Theorem 4.1 with $\mu = 1$,

$$e^{2\tau} \|\Phi_\tau w_0\|_m^2 \leq C_m^2 \|w_0\|_m^2 \qquad (\tau \geq 0),$$

so the supremum in (4.3) is finite and the upper bound in (4.4) follows. The lower bound follows by choosing $\tau = 0$, since $\Phi_0 w_0 = w_0$.

For the contraction estimate, first observe that the solution starting from $\Phi_s w_0$ at time 0 is the same as the original solution starting from w_0 and observed after the additional time s. This is the semiflow property following from uniqueness:

$$\Phi_\tau(\Phi_s w_0) = \Phi_{\tau+s} w_0, \qquad \tau, s \geq 0.$$

Therefore

$$\mathcal{L}_m(\Phi_s w_0) = \sup_{\tau \geq 0} e^{2\tau} \|\Phi_{\tau+s} w_0\|_m^2.$$

Writing $\sigma = \tau + s$, the range $\tau \geq 0$ becomes $\sigma \geq s$, and also $e^{2\tau} = e^{2(\sigma - s)} = e^{-2s} e^{2\sigma}$. Hence

$$\mathcal{L}_m(\Phi_s w_0) = e^{-2s} \sup_{\sigma \geq s} e^{2\sigma} \|\Phi_\sigma w_0\|_m^2 \leq e^{-2s} \mathcal{L}_m(w_0).$$

□

Theorem 4.4 (Closure of the perturbative weighted branch)**.** *For every $m > \frac{7}{2}$, the perturbative weighted ancient branch consists only of the zero solution.*

Proof. Let w be an ancient solution in the branch and fix $\tau_0 \in \mathbb{R}$. For any $s < \tau_0$, the time-shifted function

$$W_s(\theta) = w(s+\theta), \qquad \theta \geq 0,$$

is a global mild solution of (4.1) with initial datum $W_s(0) = w(s)$. Since $\|w(s)\|_m \leq \rho_m$, Theorem 4.1 applies to this datum. By uniqueness in $C([0,\infty); L^2_\sigma(m))$, this shifted ancient solution agrees with the semiflow solution:

$$W_s(\theta) = \Phi_\theta w(s), \qquad \theta \geq 0.$$

Choosing $\theta = \tau_0 - s$ gives

$$w(\tau_0) = \Phi_{\tau_0 - s} w(s).$$

Applying Theorem 4.3,

$$\mathcal{L}_m(w(\tau_0)) \leq e^{-2(\tau_0 - s)} \mathcal{L}_m(w(s)) \leq C_m^2 e^{-2(\tau_0 - s)} \|w(s)\|_m^2 \leq C_m^2 \rho_m^2 e^{-2(\tau_0 - s)}.$$

The left-hand side is nonnegative and independent of s. As $s \to -\infty$, the factor $e^{-2(\tau_0 - s)}$ tends to zero. Therefore $\mathcal{L}_m(w(\tau_0)) = 0$. The coercive lower bound (4.4) gives

$$0 \leq \|w(\tau_0)\|_m^2 \leq \mathcal{L}_m(w(\tau_0)) = 0,$$

so $w(\tau_0) = 0$ in $L^2_\sigma(m)$. Since τ_0 was arbitrary, $w \equiv 0$ on $\mathbb{R}$. $\square$

Remark 4.5. This theorem closes only the perturbative weighted branch near zero vorticity. It does not exclude a nonperturbative weighted ancient solution lying outside the Gallay–Wayne small-data ball.

5. The Burgers-vortex model branch

The next Lyapunov construction applies to the strained Burgers-vortex model. It is a rigorous model theorem, not a theorem for the unforced equation (1.1).

5.1. **The strained vorticity equation.** Let $x = (x_h, x_3) \in \mathbb{R}^2 \times \mathbb{R}$ and let

$$A = \operatorname{diag}\left(-\frac{1}{2}, -\frac{1}{2}, 1\right).$$

The Burgers-vortex vorticity equation is

$$\partial_\tau \Omega + (U \cdot \nabla)\Omega - (\Omega \cdot \nabla)U = \Delta\Omega - (Ax \cdot \nabla)\Omega + A\Omega, \qquad \nabla \cdot \Omega = 0, \tag{5.1}$$

with U recovered from Ω by the Biot–Savart law. It has the stationary family

$$\Omega = \alpha G, \qquad G(x) = \begin{pmatrix} 0 \\ 0 \\ g(x_h) \end{pmatrix}, \qquad g(x_h) = \frac{1}{4\pi} e^{-|x_h|^2/4}, \qquad \alpha \in \mathbb{R}. \tag{5.2}$$

For $m > 1$, let $L^2_h(m)$ denote the horizontal weighted space on $\mathbb{R}^2$ with norm

$$\|f\|^2_{L^2_h(m)} = \int_{\mathbb{R}^2} |f(x_h)|^2 \left(1 + \frac{|x_h|^2}{4m}\right)^m dx_h.$$

Set

$$X(m) = BC(\mathbb{R}_{x_3}; L^2_h(m)), \qquad X_0(m) = \left\{ f \in X(m) : \int_{\mathbb{R}^2} f(x_h, x_3)\, dx_h = 0 \text{ for every } x_3 \right\},$$

and

$$\mathcal{X}_{\text{full}}(m) = X(m)^3, \qquad \mathcal{X}_{\text{mod}}(m) = X(m) \times X(m) \times X_0(m).$$

Definition 5.1 (Admissible Burgers-vortex solutions)**.** In the Burgers-vortex section, a mild solution $\Omega : \mathbb{R} \to \mathcal{X}_{\mathrm{full}}(m)$ is called admissible if the following operations are justified for every compact time interval:

(i) horizontal integrations by parts against constants and against x_h are valid, with no boundary contribution at $|x_h| = \infty$;

(ii) the identity
$$\int_{\mathbb{R}^2} \big[(U \cdot \nabla)\Omega - (\Omega \cdot \nabla)U\big]_3 \, dx_h = 0$$
holds in distributions in x_3 and τ;

(iii) the map
$$\tau \mapsto \int_{\mathbb{R}^2} \Omega_3(x_h, x_3, \tau) \, dx_h$$
is absolutely continuous after testing in x_3.

Smooth rapidly decaying solutions are admissible. The Gallay–Maekawa mild solutions used below are assumed in this admissible class; this is precisely the amount of regularity needed for the circulation argument.

Lemma 5.2 (Circulation coordinate)**.** *Let $m > 1$ and let $\Omega \in \mathcal{X}_{\mathrm{full}}(m)$ be divergence-free. Assume that the horizontal integration by parts*
$$\int_{\mathbb{R}^2} \nabla_h \cdot \Omega_h(x_h, x_3) \, dx_h = 0$$
is valid for every x_3, or in the sense of distributions in x_3. Then
$$\alpha[\Omega](x_3) = \int_{\mathbb{R}^2} \Omega_3(x_h, x_3) \, dx_h$$
is well-defined and independent of x_3. Writing the common value as $\alpha[\Omega]$, one has
$$Q\Omega := \Omega - \alpha[\Omega] G \in \mathcal{X}_{\mathrm{mod}}(m).$$

Proof. Since $m > 1$, the horizontal weighted space embeds into $L^1(\mathbb{R}^2)$. Indeed, by Cauchy–Schwarz,
$$\int_{\mathbb{R}^2} |f(x_h)| \, dx_h \le \left(\int_{\mathbb{R}^2} |f(x_h)|^2 \left(1 + \frac{|x_h|^2}{4m}\right)^m dx_h \right)^{1/2} \left(\int_{\mathbb{R}^2} \left(1 + \frac{|x_h|^2}{4m}\right)^{-m} dx_h \right)^{1/2}.$$
The second integral is finite because the dimension is two and $m > 1$. Hence $\alpha[\Omega](x_3)$ is well-defined.

Using $\nabla \cdot \Omega = 0$ gives
$$\partial_{x_3} \alpha[\Omega](x_3) = \int_{\mathbb{R}^2} \partial_{x_3} \Omega_3 \, dx_h = - \int_{\mathbb{R}^2} \nabla_h \cdot \Omega_h \, dx_h = 0,$$
where the last equality is the stated horizontal integration-by-parts assumption. Thus $\partial_{x_3} \alpha[\Omega] = 0$, classically or distributionally according to the interpretation of the previous display, and $\alpha[\Omega]$ is independent of x_3.

Finally,
$$\int_{\mathbb{R}^2} g(x_h) \, dx_h = \frac{1}{4\pi} \int_0^\infty e^{-r^2/4} \, 2\pi r \, dr = \frac{1}{2} \int_0^\infty e^{-r^2/4} r \, dr = 1.$$
Therefore the third component of $Q\Omega = \Omega - \alpha[\Omega] G$ has zero horizontal mean for every x_3, while the first two components are unchanged. Hence $Q\Omega \in \mathcal{X}_{\mathrm{mod}}(m)$. □

Lemma 5.3 (Conservation of circulation). *Let $m > 1$, and let $\Omega(\tau)$ be an admissible mild solution of (5.1) with values in $\mathcal{X}_{\mathrm{full}}(m)$. Then $\alpha[\Omega(\tau)]$ is independent of τ. If $\bar{\alpha}$ denotes this constant, then*

$$\omega(\tau) = \Omega(\tau) - \bar{\alpha}G$$

solves the perturbation equation around the fixed Burgers vortex $\bar{\alpha}G$ and belongs to $\mathcal{X}_{\mathrm{mod}}(m)$ for every τ.

Proof. By admissibility, the following horizontal integrations can be read either classically for smooth rapidly decaying solutions or distributionally for admissible mild solutions. Integrating the third component of (5.1) over x_h, the nonlinear term does not contribute because

$$(U \cdot \nabla)\Omega - (\Omega \cdot \nabla)U = -\nabla \times (U \times \Omega),$$

where the identity follows from

$$\nabla \times (U \times \Omega) = (\Omega \cdot \nabla)U - (U \cdot \nabla)\Omega + U(\nabla \cdot \Omega) - \Omega(\nabla \cdot U),$$

and both U and Ω are divergence-free. The third component of a curl is

$$[\nabla \times (U \times \Omega)]_3 = \partial_{x_1}(U \times \Omega)_2 - \partial_{x_2}(U \times \Omega)_1,$$

a horizontal divergence. Its integral over $\mathbb{R}^2_{x_h}$ vanishes by the horizontal decay.

For the linear part,

$$\left[\Delta\Omega - (Ax \cdot \nabla)\Omega + A\Omega\right]_3 = \Delta_h\Omega_3 + \partial_{x_3}^2\Omega_3 + \frac{1}{2}x_h \cdot \nabla_h\Omega_3 - x_3\partial_{x_3}\Omega_3 + \Omega_3.$$

After horizontal integration, the Δ_h term vanishes and

$$\int_{\mathbb{R}^2} \frac{1}{2}x_h \cdot \nabla_h\Omega_3 \, dx_h = -\int_{\mathbb{R}^2} \Omega_3 \, dx_h.$$

Here the equality is obtained from

$$\nabla_h \cdot (x_h\Omega_3) = 2\Omega_3 + x_h \cdot \nabla_h\Omega_3$$

and integration over $\mathbb{R}^2$. The last displayed term cancels with the horizontal integral of $+\Omega_3$. Hence

$$\frac{d}{d\tau}\alpha[\Omega(\tau)] = \partial_{x_3}^2\alpha[\Omega(\tau)] - x_3\partial_{x_3}\alpha[\Omega(\tau)].$$

By Theorem 5.2, the right-hand side is zero. Thus the circulation is conserved. Subtracting the stationary solution $\bar{\alpha}G$ from (5.1) gives the fixed-circulation perturbation equation, and Theorem 5.2 gives membership in $\mathcal{X}_{\mathrm{mod}}(m)$. □

Theorem 5.4 (Gallay–Maekawa stability). *Let $m > 2$ and $\alpha \in \mathbb{R}$. There exist constants $\delta_{\alpha,m} > 0$ and $C_{\alpha,m} \geq 1$ such that every divergence-free initial perturbation*

$$\omega_0 \in \mathcal{X}_{\mathrm{mod}}(m), \qquad \|\omega_0\|_{\mathcal{X}_{\mathrm{mod}}(m)} \leq \delta_{\alpha,m},$$

generates a unique global mild solution of the fixed-circulation perturbation equation around αG, denoted $\Phi_\tau^\alpha\omega_0$, and

$$\|\Phi_\tau^\alpha\omega_0\|_{\mathcal{X}_{\mathrm{mod}}(m)} \leq C_{\alpha,m}e^{-\tau/2}\|\omega_0\|_{\mathcal{X}_{\mathrm{mod}}(m)} \qquad (\tau \geq 0).$$

Proof. This is the Gallay–Maekawa nonlinear stability theorem for Burgers vortices in the horizontal weighted space [6]. It is used as an external input. The only conclusions used below are existence, uniqueness, the fixed-circulation semiflow property, and the displayed exponential decay. □

Proposition 5.5 (Profile-relative Lyapunov functional)**.** *Fix $m > 2$ and $\alpha \in \mathbb{R}$. On the ball*

$$\|\omega_0\|_{\mathcal{X}_{\mathrm{mod}}(m)} \le \delta_{\alpha,m}$$

define

$$\mathcal{B}_{\alpha,m}(\omega_0) = \sup_{\tau \ge 0} e^{\tau} \|\Phi^{\alpha}_{\tau}\omega_0\|^2_{\mathcal{X}_{\mathrm{mod}}(m)}.$$

Then

$$\|\omega_0\|^2_{\mathcal{X}_{\mathrm{mod}}(m)} \le \mathcal{B}_{\alpha,m}(\omega_0) \le C^2_{\alpha,m}\|\omega_0\|^2_{\mathcal{X}_{\mathrm{mod}}(m)},$$

and

$$\mathcal{B}_{\alpha,m}(\Phi^{\alpha}_{s}\omega_0) \le e^{-s}\mathcal{B}_{\alpha,m}(\omega_0) \qquad (s \ge 0).$$

Proof. Gallay–Maekawa stability gives, for every $\tau \ge 0$,

$$e^{\tau}\|\Phi^{\alpha}_{\tau}\omega_0\|^2_{\mathcal{X}_{\mathrm{mod}}(m)} \le C^2_{\alpha,m}\|\omega_0\|^2_{\mathcal{X}_{\mathrm{mod}}(m)}.$$

Taking the supremum over $\tau \ge 0$ proves that $\mathcal{B}_{\alpha,m}$ is finite and gives the upper coercive bound. The lower bound follows from the value at $\tau = 0$:

$$\mathcal{B}_{\alpha,m}(\omega_0) \ge \|\Phi^{\alpha}_{0}\omega_0\|^2_{\mathcal{X}_{\mathrm{mod}}(m)} = \|\omega_0\|^2_{\mathcal{X}_{\mathrm{mod}}(m)}.$$

For $s \ge 0$, uniqueness of the fixed-circulation perturbation equation gives the semiflow identity

$$\Phi^{\alpha}_{\tau}(\Phi^{\alpha}_{s}\omega_0) = \Phi^{\alpha}_{\tau+s}\omega_0.$$

Consequently,

$$\mathcal{B}_{\alpha,m}(\Phi^{\alpha}_{s}\omega_0) = \sup_{\tau \ge 0} e^{\tau}\|\Phi^{\alpha}_{\tau+s}\omega_0\|^2_{\mathcal{X}_{\mathrm{mod}}(m)}.$$

Writing $\sigma = \tau + s$, one gets

$$\mathcal{B}_{\alpha,m}(\Phi^{\alpha}_{s}\omega_0) = e^{-s}\sup_{\sigma \ge s} e^{\sigma}\|\Phi^{\alpha}_{\sigma}\omega_0\|^2_{\mathcal{X}_{\mathrm{mod}}(m)} \le e^{-s}\mathcal{B}_{\alpha,m}(\omega_0).$$

□

Definition 5.6 (Burgers-vortex ancient tube)**.** Fix $m > 2$. The Burgers-vortex ancient tube consists of all admissible ancient mild solutions $\Omega : \mathbb{R} \to \mathcal{X}_{\mathrm{full}}(m)$ of (5.1) such that, with $\bar{\alpha} = \alpha[\Omega(\tau)]$,

$$\sup_{\tau \in \mathbb{R}} \|\Omega(\tau) - \bar{\alpha}G\|_{\mathcal{X}_{\mathrm{mod}}(m)} \le \delta_{\bar{\alpha},m}.$$

The phrase "ancient mild" also includes the compatibility that for every $s \in \mathbb{R}$, the shifted perturbation

$$\theta \mapsto \Omega(s+\theta) - \bar{\alpha}G, \qquad \theta \ge 0,$$

is the Gallay–Maekawa fixed-circulation mild solution launched from $\Omega(s) - \bar{\alpha}G$.

Theorem 5.7 (Ancient rigidity in the Burgers-vortex tube)**.** *Let $m > 2$. If Ω belongs to the Burgers-vortex ancient tube, then*

$$\Omega(\tau) \equiv \bar{\alpha}G \qquad \text{for all } \tau \in \mathbb{R},$$

where $\bar{\alpha} = \alpha[\Omega(\tau)]$.

Proof. By Theorem 5.3, $\bar{\alpha}$ is independent of time and

$$\omega(\tau) = \Omega(\tau) - \bar{\alpha}G$$

is an ancient solution of the fixed-circulation perturbation equation in the small ball of $\mathcal{X}_{\mathrm{mod}}(m)$. Fix $\tau_0 \in \mathbb{R}$ and $s < \tau_0$. The shifted function

$$\omega_s(\theta) = \omega(s+\theta), \qquad \theta \ge 0,$$

is a global mild solution of the fixed-circulation perturbation equation with initial datum $\omega_s(0) = \omega(s)$. The tube hypothesis gives

$$\|\omega(s)\|_{\mathcal{X}_{\mathrm{mod}}(m)} \le \delta_{\bar\alpha,m},$$

so Gallay–Maekawa uniqueness applies. Therefore

$$\omega(\tau_0) = \Phi^{\bar\alpha}_{\tau_0 - s}\omega(s).$$

Using Theorem 5.5,

$$\mathcal{B}_{\bar\alpha,m}(\omega(\tau_0)) \le e^{-(\tau_0-s)}\mathcal{B}_{\bar\alpha,m}(\omega(s)) \le e^{-(\tau_0-s)}C^2_{\bar\alpha,m}\|\omega(s)\|^2_{\mathcal{X}_{\mathrm{mod}}(m)}.$$

The tube bound makes the last norm at most $\delta^2_{\bar\alpha,m}$, uniformly in s. Letting $s \to -\infty$ gives $\mathcal{B}_{\bar\alpha,m}(\omega(\tau_0)) = 0$. By the lower coercive bound in Theorem 5.5,

$$0 \le \|\omega(\tau_0)\|^2_{\mathcal{X}_{\mathrm{mod}}(m)} \le \mathcal{B}_{\bar\alpha,m}(\omega(\tau_0)) = 0.$$

Thus $\omega(\tau_0) = 0$. Since τ_0 was arbitrary, $\Omega(\tau) = \bar\alpha G$ for every τ. □

Assumption 5.8 (Reduction to the Burgers-vortex model)**.** Let $\mathcal{C}$ be a class of ancient solutions of an unforced or renormalized Navier–Stokes problem. Assume there are $m > 2$, a map

$$\mathcal{T} : \mathcal{C} \to C_{\mathrm{loc}}(\mathbb{R}; \mathcal{X}_{\mathrm{full}}(m)),$$

and a nonnegative defect $d[u(\tau)]$ such that for every $u \in \mathcal{C}$:

(i) $\Omega = \mathcal{T}[u]$ is an admissible ancient mild solution of (5.1) in the sense of Theorem 5.1;

(ii) with $\bar\alpha = \alpha[\Omega]$,

$$\|\Omega(\tau) - \bar\alpha G\|_{\mathcal{X}_{\mathrm{mod}}(m)} \le C_* d[u(\tau)] \qquad (\tau \in \mathbb{R});$$

(iii) $\sup_{\tau\in\mathbb{R}} d[u(\tau)] \le \delta_{\bar\alpha,m}/C_*$.

Corollary 5.9 (Conditional model import)**.** *Under Theorem 5.8, every $u \in \mathcal{C}$ is mapped by $\mathcal{T}$ to an exact Burgers vortex:*

$$\mathcal{T}[u](\tau) \equiv \bar\alpha G.$$

Proof. Let $\Omega = \mathcal{T}[u]$. By assumption (i), Ω is an ancient mild solution of (5.1). By Theorem 5.3, its circulation $\bar\alpha = \alpha[\Omega]$ is constant in time. Assumptions (ii) and (iii) give, for every $\tau \in \mathbb{R}$,

$$\|\Omega(\tau) - \bar\alpha G\|_{\mathcal{X}_{\mathrm{mod}}(m)} \le C_* d[u(\tau)] \le \delta_{\bar\alpha,m}.$$

Thus Ω belongs to the Burgers-vortex ancient tube. Applying Theorem 5.7 gives $\Omega(\tau) \equiv \bar\alpha G$, which is the claimed conclusion. □

Remark 5.10. Theorem 5.9 is intentionally conditional. The strained equation (5.1) contains the imposed axial strain A, whereas the unforced Navier–Stokes equations do not. A separate analytic reduction is needed before Burgers-vortex rigidity can be used for an unforced singularity core.

6. The scalar Γ defect

We now turn to the remaining axisymmetric bounded-swirl branch. The rigorous output of this section is an exact scalar identity and a precise shell-control criterion. The criterion is not known to hold in full generality.

6.1. The equation for Γ.

Lemma 6.1 (Swirl equation)**.** *Let (u,p) be a smooth axisymmetric solution of (1.1). Then $\Gamma = ru_\theta$ satisfies*

$$\partial_t\Gamma + b\cdot\nabla\Gamma + \frac{2}{r}\partial_r\Gamma = \Delta\Gamma, \qquad b = u_re_r + u_ze_z. \tag{6.1}$$

Proof. For an axisymmetric vector field, the vector Laplacian has angular component

$$(\Delta u)_\theta = \left(\partial_r^2 + \frac{1}{r}\partial_r + \partial_z^2 - \frac{1}{r^2}\right)u_\theta,$$

and the convective derivative has angular component

$$[(u\cdot\nabla)u]_\theta = u_r\partial_ru_\theta + u_z\partial_zu_\theta + \frac{u_r}{r}u_\theta.$$

The pressure has no angular component because the flow is axisymmetric: $(\nabla p)_\theta = r^{-1}\partial_\theta p = 0$. Therefore the angular component of (1.1) is

$$\partial_tu_\theta + u_r\partial_ru_\theta + u_z\partial_zu_\theta + \frac{u_r}{r}u_\theta = \left(\partial_r^2 + \frac{1}{r}\partial_r + \partial_z^2 - \frac{1}{r^2}\right)u_\theta.$$

Multiplying by r and using

$$r\partial_ru_\theta = \partial_r(ru_\theta) - u_\theta = \partial_r\Gamma - u_\theta$$

gives cancellation of the zeroth-order transport term:

$$ru_r\partial_ru_\theta + u_ru_\theta = u_r\partial_r\Gamma.$$

For the diffusion,

$$r\left(\partial_r^2 + \frac{1}{r}\partial_r - \frac{1}{r^2}\right)u_\theta = \partial_r^2\Gamma - \frac{1}{r}\partial_r\Gamma.$$

Indeed, since $u_\theta = \Gamma/r$,

$$\partial_ru_\theta = \frac{1}{r}\partial_r\Gamma - \frac{1}{r^2}\Gamma$$

and

$$\partial_r^2u_\theta = \frac{1}{r}\partial_r^2\Gamma - \frac{2}{r^2}\partial_r\Gamma + \frac{2}{r^3}\Gamma.$$

Thus

$$r\partial_r^2u_\theta + \partial_ru_\theta - \frac{1}{r}u_\theta = \partial_r^2\Gamma - \frac{1}{r}\partial_r\Gamma.$$

Since

$$\Delta\Gamma = \partial_r^2\Gamma + \frac{1}{r}\partial_r\Gamma + \partial_z^2\Gamma$$

for axisymmetric scalar functions, the radial diffusion term is

$$\partial_r^2\Gamma - \frac{1}{r}\partial_r\Gamma + \partial_z^2\Gamma = \Delta\Gamma - \frac{2}{r}\partial_r\Gamma.$$

Combining the displayed identities gives (6.1). □

Proposition 6.2 (Weighted scalar defect identity)**.** *Let u be a smooth axisymmetric solution of (1.1). Set $W = \Gamma$, and let $\phi = \phi(r,z,t) \geq 0$ be a smooth compactly supported axisymmetric weight. Then*

$$\frac{1}{2}\frac{d}{dt}\int_{\mathbb{R}^3} W^2\phi\,dx + \int_{\mathbb{R}^3} |\nabla W|^2\phi\,dx = \frac{1}{2}\int_{\mathbb{R}^3} W^2\left(\partial_t\phi + \Delta\phi + b\cdot\nabla\phi + \frac{2}{r}\partial_r\phi\right)dx. \tag{6.2}$$

If $W = \Gamma - c$, the same identity holds provided either $c = 0$ or ϕ vanishes in a neighbourhood of the axis. Without this restriction, the right-hand side in (6.2) must be augmented by the axis boundary term

$$2\pi c^2 \int_{\mathbb{R}} \phi(0, z, t)\, dz.$$

Proof. Since $W = \Gamma$, Theorem 6.1 gives

$$\partial_t W + b \cdot \nabla W + \frac{2}{r}\partial_r W = \Delta W.$$

Equivalently,

$$\partial_t W = \Delta W - b \cdot \nabla W - \frac{2}{r}\partial_r W.$$

Multiply by $W\phi$ and integrate over $\mathbb{R}^3$. We compute each term separately. The time derivative gives

$$\int W \partial_t W\, \phi\, dx = \frac{1}{2}\frac{d}{dt}\int W^2 \phi\, dx - \frac{1}{2}\int W^2 \partial_t \phi\, dx.$$

For the Laplacian,

$$\int W \Delta W\, \phi\, dx = -\int |\nabla W|^2 \phi\, dx - \int W \nabla W \cdot \nabla \phi\, dx.$$

Since $W\nabla W = \frac{1}{2}\nabla(W^2)$, compact support of ϕ gives

$$-\int W \nabla W \cdot \nabla \phi\, dx = -\frac{1}{2}\int \nabla(W^2) \cdot \nabla \phi\, dx = \frac{1}{2}\int W^2 \Delta \phi\, dx.$$

Because u is divergence-free and b is the meridional part of an axisymmetric divergence-free vector field,

$$\nabla \cdot b = 0$$

in the full three-dimensional divergence sense. Explicitly,

$$\nabla \cdot b = \frac{1}{r}\partial_r(r u_r) + \partial_z u_z = \nabla \cdot u = 0.$$

Hence

$$-\int W (b \cdot \nabla W)\phi\, dx = -\frac{1}{2}\int b \cdot \nabla(W^2)\phi\, dx = \frac{1}{2}\int W^2 b \cdot \nabla \phi\, dx.$$

Finally,

$$-\int W \frac{2}{r}\partial_r W\, \phi\, dx = -\int \frac{1}{r}\partial_r(W^2)\phi\, dx.$$

In cylindrical variables $dx = r\, dr\, d\theta\, dz$, this term equals

$$-\int_0^{2\pi}\int_{\mathbb{R}}\int_0^\infty \partial_r(W^2)\phi\, dr\, dz\, d\theta.$$

There is no boundary contribution at $r = \infty$, because ϕ is compactly supported. At the axis, one can first insert a smooth radial cutoff which vanishes for $r < \delta$, perform the integration by parts on $\delta < r < \infty$, and then let $\delta \downarrow 0$. Smoothness of a bounded axisymmetric velocity gives $u_\theta(r, z, t) = O(r)$ near $r = 0$; hence $\Gamma(r, z, t) = r u_\theta(r, z, t) = O(r^2)$. Therefore $W^2\phi = \Gamma^2\phi = O(r^4)$ at the axis, and the boundary term converges to zero. Thus

$$-\int_0^{2\pi}\int_{\mathbb{R}}\int_0^\infty \partial_r(W^2)\phi\, dr\, dz\, d\theta = \int_0^{2\pi}\int_{\mathbb{R}}\int_0^\infty W^2 \partial_r \phi\, dr\, dz\, d\theta = \int W^2 \frac{1}{r}\partial_r \phi\, dx.$$

Therefore

$$-\int W\frac{2}{r}\partial_r W\,\phi\,dx = \int W^2\frac{1}{r}\partial_r\phi\,dx = \frac{1}{2}\int W^2\frac{2}{r}\partial_r\phi\,dx.$$

Putting the time, diffusion, transport, and singular radial terms together gives (6.2).

If $W = \Gamma - c$, the differential equation is unchanged because derivatives of c vanish. The only altered step is the radial integration by parts at the axis. Since $\Gamma(0, z, t) = 0$, one has $W(0, z, t) = -c$, and the missing boundary contribution is

$$\int_0^{2\pi}\int_{\mathbb{R}} c^2\phi(0, z, t)\,dz\,d\theta = 2\pi c^2\int_{\mathbb{R}}\phi(0, z, t)\,dz.$$

This term vanishes if $c = 0$ or if ϕ is supported away from the axis. □

Remark 6.3. For bounded ancient mild axisymmetric solutions with bounded Γ, this identity is not invoked automatically. It is invoked only under Theorem 6.6, where smoothness, finiteness of the integrals, and validity of the weighted identities for the chosen weights are assumed explicitly. No vortex-stretching term appears in (6.2); the nonlinearity enters only through the drift $b \cdot \nabla\phi$.

6.2. **Anisotropic shell weights.**

Let

$$\phi_{R,Z,\zeta}(r, z, t) = \lambda_R(r)\eta_Z(z - \zeta(t)),$$

where λ_R, η_Z are smooth cutoffs and ζ is a smooth axial modulation. Write $\eta_Z = \eta_Z(z - \zeta(t))$ in formulas below.

Proposition 6.4 (Shell-error decomposition)**.** *Let $W = \Gamma$. With the notation above,*

$$\frac{1}{2}\frac{d}{dt}\int W^2\lambda_R\eta_Z\,dx + \int|\nabla W|^2\lambda_R\eta_Z\,dx = \frac{1}{2}\int W^2\mathcal{E}_{R,Z,\zeta}[u]\,dx, \tag{6.3}$$

where

$$\mathcal{E}_{R,Z,\zeta}[u] = \eta_Z\left(\lambda_R'' + \frac{3}{r}\lambda_R' + u_r\lambda_R'\right) + \lambda_R\left(\eta_Z'' + (u_z - \dot\zeta)\eta_Z'\right). \tag{6.4}$$

Proof. For an axisymmetric scalar,

$$\Delta(\lambda_R\eta_Z) = \left(\lambda_R'' + \frac{1}{r}\lambda_R'\right)\eta_Z + \lambda_R\eta_Z''.$$

This is the scalar cylindrical Laplacian

$$\Delta f = \partial_r^2 f + \frac{1}{r}\partial_r f + \partial_z^2 f$$

for functions independent of θ, applied to the separated product $\lambda_R(r)\eta_Z(z - \zeta(t))$. There are no mixed derivatives because λ_R has no z-dependence and η_Z has no r-dependence. Also

$$\frac{2}{r}\partial_r(\lambda_R\eta_Z) = \frac{2}{r}\lambda_R'\eta_Z,$$

and

$$b\cdot\nabla(\lambda_R\eta_Z) = u_r\lambda_R'\eta_Z + u_z\lambda_R\eta_Z'.$$

Since the axial argument is $z - \zeta(t)$,

$$\partial_t(\lambda_R\eta_Z) = -\dot\zeta\,\lambda_R\eta_Z'.$$

Adding these four contributions gives

$$\partial_t\phi + \Delta\phi + b\cdot\nabla\phi + \frac{2}{r}\partial_r\phi = \eta_Z\left(\lambda_R'' + \frac{3}{r}\lambda_R' + u_r\lambda_R'\right) + \lambda_R\left(\eta_Z'' + (u_z - \dot\zeta)\eta_Z'\right).$$

Substitution into Theorem 6.2 proves the formula. □

Remark 6.5 (The radial geometry). The differential expression

$$\lambda_R'' + \frac{3}{r}\lambda_R'$$

is the radial Laplacian in four dimensions. Thus the natural radial barrier for the Γ-defect is adapted to $\partial_r^2 + 3r^{-1}\partial_r$. The shell errors in (6.4) are precisely radial transport through $u_r\lambda_R'$, axial drift mismatch through $(u_z - \dot\zeta)\eta_Z'$, and the two cutoff-curvature terms.

6.3. **A coercive shell-control criterion.** The following statement is deliberately conditional. It spells out a set of quantitative hypotheses under which the scalar defect identity forces Γ to vanish. The difficult open step is proving these hypotheses for the whole nonperiodic bounded-Γ ancient branch.

Assumption 6.6 (Coercive shell control)**.** Let u be a bounded ancient mild axisymmetric solution with bounded Γ. Assume, in addition, that u is smooth on $\mathbb{R}^3 \times (-\infty, 0)$ and that the identities Theorems 6.1, 6.2 and 6.4 hold for the weights below. There exist expanding smooth weights

$$\phi_n(r,z,t) = \lambda_n(r)\eta_n(z - \zeta_n(t)), \qquad 0 \le \phi_n \le 1, \qquad \phi_n \uparrow 1$$

locally uniformly on compact subsets of $\mathbb{R}^3 \times (-\infty, 0)$, where $\lambda_n, \eta_n, \zeta_n$ are smooth and ϕ_n is compactly supported in space for each fixed time. There are constants $a_n > 0$, $M_n < \infty$, $\varepsilon_n \ge 0$, with $\varepsilon_n/a_n \to 0$, such that

$$L_n(t) := \int_{\mathbb{R}^3} \Gamma^2 \phi_n(t)\, dx < \infty$$

for every $t < 0$, L_n is locally absolutely continuous on $(-\infty, 0)$, and for every compact interval $[s,t] \subset (-\infty, 0)$ all integrals below are finite. For all $s < t < 0$,

$$\int_{\mathbb{R}^3} |\nabla\Gamma|^2 \phi_n(\sigma)\, dx \ge a_n L_n(\sigma) \qquad \text{for a.e. } \sigma \in (s,t), \tag{6.5}$$

$$\left| \int_{\mathbb{R}^3} \Gamma^2 \mathcal{E}_n[u]\, dx \right| \le 2\varepsilon_n \qquad \text{for a.e. } \sigma \in (s,t), \tag{6.6}$$

and

$$\sup_{\sigma < t} L_n(\sigma) \le M_n \qquad \text{for every fixed } t < 0. \tag{6.7}$$

Here $\mathcal{E}_n[u]$ is the shell-error density (6.4) associated with ϕ_n.

Theorem 6.7 (Conditional scalar flattening)**.** *Let u be a bounded ancient mild axisymmetric solution with bounded* Γ*. If Theorem 6.6 holds, then*

$$\Gamma(x,t) = 0 \qquad \textit{for every } (x,t) \in \mathbb{R}^3 \times (-\infty, 0).$$

Proof. Fix n and abbreviate $W = \Gamma$. Integrating (6.3) from s to t, using (6.5) and (6.6), gives

$$\frac{1}{2}L_n(t) - \frac{1}{2}L_n(s) + a_n \int_s^t L_n(\sigma)\, d\sigma \le \varepsilon_n(t-s).$$

Indeed, the shell identity gives

$$\frac{1}{2}L_n(t) - \frac{1}{2}L_n(s) + \int_s^t \int_{\mathbb{R}^3} |\nabla\Gamma|^2 \phi_n\, dx\, d\sigma = \frac{1}{2}\int_s^t \int_{\mathbb{R}^3} \Gamma^2 \mathcal{E}_n[u]\, dx\, d\sigma.$$

The coercivity assumption bounds the dissipation from below by

$$\int_s^t \int_{\mathbb{R}^3} |\nabla\Gamma|^2 \phi_n \, dx \, d\sigma \geq a_n \int_s^t L_n(\sigma) \, d\sigma,$$

and the error assumption bounds the right-hand side above by $\varepsilon_n(t-s)$.

Since L_n is locally absolutely continuous by Theorem 6.6, the preceding integral inequality is equivalent to the differential inequality, for a.e. $t < 0$,

$$\frac{d}{dt} L_n(t) + 2a_n L_n(t) \leq 2\varepsilon_n.$$

To see this explicitly, apply the integral inequality on $(t, t+h)$, divide by h, and let $h \downarrow 0$ at a Lebesgue point of L_n' and L_n. Multiplying by the integrating factor $e^{2a_n t}$ gives, for a.e. t,

$$\frac{d}{dt}\left(e^{2a_n t} L_n(t)\right) = e^{2a_n t}\left(\frac{d}{dt} L_n(t) + 2a_n L_n(t)\right) \leq 2\varepsilon_n e^{2a_n t}.$$

Integrating this inequality from s to t yields

$$L_n(t) \leq e^{-2a_n(t-s)} L_n(s) + \frac{\varepsilon_n}{a_n}\left(1 - e^{-2a_n(t-s)}\right).$$

By (6.7), $L_n(s) \leq M_n$ for $s < t$. Letting $s \to -\infty$ gives

$$L_n(t) \leq \frac{\varepsilon_n}{a_n}.$$

Now let $n \to \infty$. Since $\varepsilon_n / a_n \to 0$ and $\phi_n \uparrow 1$ locally uniformly, for every compact set $K \subset \mathbb{R}^3$,

$$\int_K \Gamma(x,t)^2 \, dx \leq \liminf_{n\to\infty} L_n(t) = 0$$

for every $t < 0$ outside a null set. The inequality follows because, for each compact K, local uniform convergence $\phi_n \to 1$ gives $\inf_K \phi_n(t) \to 1$, and hence

$$\int_K \Gamma^2 \, dx = \lim_{n\to\infty} \int_K \Gamma^2 \phi_n \, dx \leq \liminf_{n\to\infty} L_n(t).$$

Thus $\Gamma(\cdot, t) = 0$ locally in L^2 for a.e. t. By the smoothness included in Theorem 6.6, Γ is continuous on $\mathbb{R}^3 \times (-\infty, 0)$. Let $t_0 < 0$ be arbitrary, and choose times $t_j \to t_0$ outside the null set for which $\Gamma(\cdot, t_j) = 0$ locally in L^2_x. For each j, continuity in space implies $\Gamma(x, t_j) = 0$ for every x: otherwise nonzero value at one point would give nonzero L^2-mass on a small ball. Passing to the limit $j \to \infty$ and using continuity in time gives $\Gamma(x, t_0) = 0$ for every $x \in \mathbb{R}^3$. Since t_0 was arbitrary, $\Gamma \equiv 0$ pointwise. ☐

Corollary 6.8 (Reduction after scalar flattening)**.** *Let u be as in Theorem 6.7. If in addition u belongs to a class for which $\Gamma \equiv 0$ implies one of the Liouville alternatives in Theorem 3.9, then u is Galilean-trivial.*

Proof. Theorem 6.7 gives $\Gamma \equiv 0$. The additional assumption places the solution in one of the already closed axisymmetric branches. The conclusion follows from Theorem 3.9. ☐

Remark 6.9. Theorem 6.7 is not an unconditional closure of the nonperiodic bounded-Γ branch. It shows exactly what must be proved: one needs shell weights for which the radial transport, axial drift mismatch, and cutoff-curvature errors are small compared with the local coercivity constant. Without such control, the scalar identity remains a structural identity rather than a Liouville theorem.

7. Summary of closed and open branches

The rigorous exclusions obtained above are the following.

Theorem 7.1 (Branch exclusion ledger)**.** *The following ancient branches are closed.*

(i) *Bounded ancient axisymmetric unforced solutions satisfying any of the four alternatives in Theorem 3.9 are Galilean-trivial.*
(ii) *The perturbative weighted-vorticity branch of Theorem 4.2 is zero for every* $m > \frac{7}{2}$.
(iii) *The Burgers-vortex ancient tube of Theorem 5.6 consists exactly of the stationary Burgers vortices* αG.
(iv) *Any bounded ancient axisymmetric solution satisfying the coercive shell-control hypotheses of Theorem 6.6, and for which* $\Gamma \equiv 0$ *enters a closed axisymmetric Liouville class, is Galilean-trivial.*

Proof. Item (i) is Theorem 3.9. Item (ii) is Theorem 4.4. Item (iii) is Theorem 5.7. Item (iv) is Theorem 6.8. □

8. Analytic dependencies

For clarity we record which assertions are proved in the paper and which are imported as external PDE inputs. This is part of the mathematical statement: none of the imported assertions is used without being named, and none of the conditional assumptions is converted into a theorem.

Proposition 8.1 (Dependency ledger)**.** *The conclusions of Theorem 7.1 depend only on the following analytic inputs.*

(i) *The axisymmetric exclusions in Theorems 3.5 to 3.8 use exactly the corresponding Liouville theorems stated in Theorems 3.1 to 3.4, together with the elementary Galilean reduction of Theorem 2.3.*
(ii) *The perturbative weighted-vorticity exclusion uses the Gallay–Wayne stability theorem stated in Theorem 4.1. Apart from that input, the proof uses only uniqueness of the weighted-vorticity semiflow, the assumed compatibility of the ancient branch with that semiflow, and the exponential decay estimate stated in Theorem 4.1.*
(iii) *The Burgers-vortex rigidity theorem uses the Gallay–Maekawa stability theorem stated in Theorem 5.4. The reduction to a fixed-circulation fiber uses the admissibility hypotheses of Theorem 5.1; in particular, the horizontal integrations by parts and the conservation of horizontal circulation are assumptions on the solution class unless they have been verified by an independent regularity and decay argument.*
(iv) *The scalar flattening result uses no Liouville theorem before* Γ *has vanished. It uses the scalar identity of Theorem 6.4, the regularity and finiteness assertions built into Theorem 6.6, the coercive estimate* (6.5)*, the shell-error estimate* (6.6)*, and the past bound* (6.7)*.*
(v) *The final Galilean-trivial conclusion after scalar flattening requires an additional inclusion of the resulting no-swirl solution in one of the Liouville classes listed in Theorem 3.9. The paper does not assert that every bounded ancient axisymmetric solution with bounded* Γ *satisfies the shell-control assumption or that every no-swirl solution belongs to a specified external Liouville class without the hypotheses of that class.*

Proof. Each item follows by tracing the cited proof. For (i), the proofs of Theorems 3.5 to 3.8 invoke only the corresponding external Liouville result before using Theorem 2.3 to remove constant vector solutions. For (ii), Theorem 4.4 first shifts the ancient solution to time s, applies the Gallay–Wayne semiflow from time s to time t, and then sends $s \to -\infty$; the only analytic estimate used in that argument is the exponential decay estimate stated in Theorem 4.1. For (iii), Theorem 5.2 and Theorem 5.3 use precisely the admissibility conditions stated in Theorem 5.1, and Theorem 5.7 then applies Theorem 5.4 on the corresponding fixed-circulation fiber. For (iv), Theorem 6.7 integrates (6.3), applies (6.5), (6.6), and (6.7), and passes to the increasing local limit $\phi_n \uparrow 1$. The final assertion is Theorem 6.8. No further compactness, classification, or regularity statement is used in these arguments. □

Remark 8.2 (Remaining bounded-swirl obstruction). The branch not closed by this paper is the full class of bounded ancient axisymmetric unforced Navier–Stokes solutions with nontrivial bounded Γ, no L^p decay of Γ, no axial periodicity, and no verified shell-control estimate of the form Theorem 6.6. The identity (6.3) reduces this obstruction to explicit transport terms, but does not eliminate them.

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